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FINAL YEAR PROJECT TITLE SUBMISSION

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PROJECT TITLE 1: AI POWERED ADDICTION RECOVERY COUNSELING PLATFORM

Introduction

Addiction to substances and behaviors is a growing public health concern, yet many people face barriers to accessing recovery support due to stigma, cost, and limited healthcare availability. Traditional counseling systems cannot meet the increasing demand for timely and continuous care. This project proposes the development of a privacy-centered, AI-powered platform that improves accessibility to addiction recovery services. The system integrates AI-driven psychological support, moderated peer communities, and professional tele-therapy into a single digital ecosystem designed to provide scalable and user-centered recovery assistance.

Problem Statement

Current addiction care models struggle with low patient engagement caused by psychosocial barriers (stigma and confidentiality concerns) and logistical challenges (cost and distance). There is no unified digital platform that combines immediate AI-based crisis support with long-term professional counseling and peer engagement. This gap limits access to sustained recovery services. The proposed project addresses this issue by creating a tiered, accessible support system that reduces entry barriers while maintaining high standards of care.

Objectives

The primary objective is to design and develop an accessible AI-powered addiction recovery platform. Specific objectives include:

- Developing a 24/7 AI counseling chatbot for emotional support and recovery guidance
- Creating moderated peer support communities
- Enabling secure connections with certified therapists
- Designing a user-friendly and secure interface
- Protecting user privacy and confidentiality
- Providing educational resources on addiction recovery

Main Functionalities

The platform offers three core services:

AI Counseling: Personalized chatbot conversations, daily check-ins, and coping strategies.

Community Support: Moderated peer groups for shared experiences and encouragement.

Professional Therapy: Secure communication with licensed therapists.

Significance of the Project

This project combines AI technology, peer support, and professional counseling to reduce barriers to addiction recovery. It promotes early intervention, continuous care, and improved mental well-being. Technologically, it demonstrates how digital innovation can address major public health challenges and strengthen community support systems.

PROJECT TITLE 2: ONLINE MENTAL HEALTH SUPPORT AND COUNSELING PLATFORM

Introduction

Mental health challenges such as stress, anxiety, and depression are increasingly affecting people of all ages, especially students and young adults. Many individuals struggle to access professional help due to stigma, lack of awareness, cost, or limited availability of services. With the widespread use of digital technology, online platforms can provide accessible and confidential mental health support. This project proposes the development of an online counseling platform that offers emotional support tools, self-help resources, and communication features to assist individuals in managing their mental well-being.

Statement of the Problem

Despite the growing need for mental health support, many people do not receive timely help. Traditional counseling services may be inaccessible due to social stigma, financial barriers, or limited availability of professionals. Additionally, individuals experiencing stress or depression often lack safe and anonymous spaces to express their feelings and seek guidance. Therefore, there is a need for a digital platform that provides accessible, private, and supportive mental health assistance.

Objectives of the Project

General Objective:

To design and develop an online platform that provides accessible mental health support and counseling services.

Specific Objectives:

- To create a user-friendly system for accessing mental health resources
- To provide anonymous communication between users and support providers
- To implement mood tracking and self-help tools for emotional management
- To offer educational content on stress management and mental wellness
- To ensure user privacy and data confidentiality

Significance of the Project

This project contributes to improving mental health awareness and support accessibility through technology. It provides individuals with a safe and convenient platform to seek help and manage emotional challenges. The system can benefit students, educators, and the wider community by promoting early support and reducing the stigma associated with mental health issues. Academically, the project demonstrates the practical application of information technology in solving real-world social problems and promoting well-being.

PROJECT TITLE 3: ETHIO TRADE ...A REAL TIME AUCTION AND BARTER SYSTEM WITH CONCURRENCY CONTROL

Introduction

The growth of digital commerce has increased the demand for online platforms that support auctions and barter transactions. However, real-time trading environments involve multiple users interacting simultaneously, which can lead to fairness and consistency issues if not properly managed. EthioTrade is a real-time web-based auction and barter system designed to ensure reliable and fair transactions. The platform integrates concurrency control mechanisms to maintain data consistency and prevent conflicts during simultaneous user interactions.

Problem Statement

In many online auction systems, simultaneous bids or transactions can cause race conditions, data inconsistencies, and unfair outcomes. Without proper synchronization and transaction control, users may experience double transactions or inaccurate results. Therefore, there is a need for a real-time trading platform that ensures fairness, prevents conflicts, and maintains consistent data under concurrent access.

Objectives

General Objective

To design and implement a real-time auction and barter system that ensures fairness and data consistency through effective concurrency control techniques.

Specific Objectives

- To develop a real-time bidding system using WebSocket technology
- To implement concurrency control mechanisms to prevent race conditions
- To ensure transaction consistency using database transaction management
- To design a structured barter request and approval workflow
- To implement secure authentication and role-based access control

Significance of the Project

Academically, this project demonstrates the practical application of concurrency control, database transaction management, and real-time system design. It allows students to implement distributed systems concepts in a real-world trading environment.

Practically, EthioTrade promotes fair and reliable digital auctions and barter exchanges by ensuring data consistency and preventing transaction conflicts. The system can serve as a foundation for secure and scalable online trading platforms.